

THE LEAGUE:

Technical Notes on a Game of University Competition under Two Institutions*

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Abstract

Four universities compete for forty school-leavers, once a year, for twenty years. Seats cost rent in both worlds, filled or not; in one world colleges set their own fees and must honour every acceptance, in the other a regulator fixes the fee and a clearing house assigns students by deferred acceptance against reported quotas. THE LEAGUE is a browser game in which a player takes over the worst of the four under either institution and tries to reach the top of a published league table. These notes set out the common environment, the two market designs, and a calibration by scripted players—revised, in this second edition, after a playtest found the scheme could be beaten by spending without thinking. It no longer can. The market kills by crowd; the scheme kills by emptiness.

1 Introduction

You have accepted the worst chair in academia. Greyfriars College is fourth of four in the league table that *The Morning Ledger* publishes every autumn; its endowment would not survive two bad roofs; its teaching is tired and its research is a rumour. The governing board, out of options and nearly out of money, asks only that you climb.

In either world each of your two departments keeps eight seats, and every seat costs rent whether a student sits in it or not. How you may climb depends on what else the world permits. In the first world you set your own fees. Applicants apply wherever they can afford, you make an offer to every applicant above a bar of your choosing, and every offer that is accepted must be honoured—the ninth student in a department, the one without a seat, costs a small fortune in emergency provision. In the second world the fee has been confiscated by a regulator and set equal for everyone. A national clearing house runs the admissions round: you report a quota and a standard, an algorithm assigns each student to the best place that will hold them, and the quota is never exceeded. The market can overflow your seats; the scheme can only leave them empty.

THE LEAGUE is a browser game in which both worlds can be played, from the same starting position, against the same opponents. These notes are its documentation, written for economists.

*The game, the simulations, and a first draft of these notes were produced in collaboration with Claude (Anthropic). The game is a single, dependency-free HTML file; it, the simulation harness, and the source of these notes live at <https://github.com/steffenhuck/ivy>. These notes are in their second edition; the revision owes its agenda to a playtest by Rustam Hakimov, whose exploit is reproduced, named, and closed in Section 5. Comments are welcome, and so are high scores.

The design brief imposed a discipline worth recording. The engine was built and calibrated headless before any interface existed: three scripted players of increasing sophistication—static, adaptive, exploitative—were run over five hundred seeded histories per institution, and parameters were tuned until the sophisticated player usually carries the worst college to the top, the adaptive player improves but rarely wins, and the static player does not improve at all. The calibration tables are reproduced in Section 6 and shipped inside the game file itself. A second discipline is pedagogical. In each world the winning strategy, discoverable by play, is a known piece of economics: in the market it is monopoly pricing over a captive segment whose demand curve must be learned by costly experiment; in the scheme it is the arithmetic of a stable mechanism—what a quota report does and does not buy (Sönmez, 1997), and what it is worth against rivals who spend rationally. The syllabus is smuggled in with the fun.

These notes are in their second edition, and the reason is a playtest. A reviewer reported reaching the top of the scheme by year five with the least interesting strategy the game admits—spend the full budget, split it equally, repeat—and holding the summit without further thought. The harness confirmed the report and sharpened it: scripted, the strategy topped the scheme in forty-three per cent of seeds, more often than the sharp player, while requiring none of the sharp player’s judgement. Section 5 records the diagnosis and the repair. The short version is that the rival colleges were hoarding cash the scheme gives them no reason to hold, and that money at the end of the game was worth nothing to anybody; both were failures of institutional imagination rather than of code, which is why they earn a section rather than a footnote.

The notes proceed as follows. Section 2 sets out the environment common to both worlds. Sections 3 and 4 specify the two institutions. Section 5 reports the playtest and the second edition’s four repairs. Section 6 reports the calibration and a comparison of the two games’ difficulty. Section 7 discusses what the comparison teaches.

2 The common environment

Universities. There are four universities, indexed i . Each carries five state variables: an endowment E^i and four qualities—research and teaching in each of two fields, STEM (S) and the humanities and social sciences (H)—written $R_S^i, T_S^i, R_H^i, T_H^i$. All money is in thousands of pounds. The league table ranks universities by the sum of the four qualities; ties are broken by endowment, then by a fixed index. The table is public. Endowments, fees, thresholds, quotas, and enrolments are not. One university is run by the player; the other three are run by heuristic personalities described in Appendix B. Starting states appear in Table 4.

Students. Each year forty fresh applicants are drawn. An applicant is a quadruple (s, b, θ, f) . The school score is $s \sim N(60, 13^2)$, truncated to $[20, 100]$. The budget is $b = 10 + 0.12(s - 60) + \varepsilon$ with $\varepsilon \sim N(0, 3.8^2)$, truncated to $[3, 25]$; money and marks travel together, mildly, and the truncation point equals the fee cap, so nothing but demand polices a high fee. The taste for research is $\theta \sim U[0, 1.2]$. The field is assigned by stratification: exactly $\text{round}(40 p_t)$ applicants prefer STEM, where $p_1 = \frac{1}{2}$ and

$$p_{t+1} = \text{clamp}\left(p_t + 0.02 \tanh\left(\frac{\bar{R}_{S,t} - \bar{R}_{H,t}}{10}\right)\right), \quad p_t \in [0.3, 0.7],$$

so that fashion drifts gently toward whichever field the sector researches better. A student attending university i in field f enjoys utility $u^i = T_f^i + \theta R_f^i$: teaching benefits everyone

alike, research prestige benefits those who care for it. Students live one year. They pay one fee, are taught once, and leave; there are no cohorts.

Quality dynamics. Investment has diminishing returns and takes effect the following year. For each quality $Q \in \{R_S, T_S, R_H, T_H\}$ with investment $I_Q \geq 0$,

$$R' = \delta R + \gamma \sqrt{I_R}, \quad T' = \delta T + \gamma \sqrt{I_T} + \kappa (\bar{s} - 60) \cdot \mathbf{1}\{\text{anyone admitted}\},$$

with decay $\delta = 0.87$, productivity $\gamma = 1.5$, and intake effect $\kappa = 0.14$, where $\bar{s}$ is the department's mean admitted score. Qualities are floored at zero. Decay is deliberately brutal: an unattended quality loses thirteen per cent a year, so standing still is a policy with consequences.

The cost of seats. Both worlds share one cost function. A department with enrolment m pays

$$C = 8\rho + c \max(0, m - 8), \quad \rho = 0.75, \quad c = 24 :$$

rent on all eight seats, filled or not, plus a steep price per student beyond them. Deferred acceptance never over-fills a quota, so the second term is reachable only in the market. Fees may exceed c (the cap is 25): overflow-for-profit is priced, not forbidden—and, as Section 6 reports, never observed.

The ledger. Fees are paid up front. With fee income F_t , the year's bills C_t —seat costs as above plus, in the second edition, the stipend bill—and investment I_t ,

$$E_{t+1} = (E_t + F_t - C_t - I_t) (1 + r), \quad r = 0.05.$$

Bankruptcy is checked after admissions and before spending: if $E_t + F_t - C_t < 0$, the college is finished. For the player this ends the game. A broke rival is frozen: it stops admitting, invests nothing, and decays.

Merit scholarships (second edition; per-head form after further playtest discussion). In either world a college sets, per field and per year, a stipend $\sigma \in [0, 20]$ per student and a qualifying bar $\beta \in [40, 95]$ in school score. Every enrolling student with $s \geq \beta$ is owed the stipend, so the bill is σ times the qualifying intake—known only after clearing, part of C_t , and therefore able to bankrupt: a generous stipend at a low bar is a road to ruin, not a bargain, and the calibration probes confirm it (Section 5). An earlier draft modelled the instrument as a committed fund whose pull was concave in its size; that version asked applicants to value a pot whose division among unknown rivals they could not compute, and it was retired for exactly that reason. The per-head form asks a student to know only her own offer.

The stipend's channel differs by world, on principle: cash nets against a price where a price exists. In the market it counts toward the fee—a qualifying student can afford the college if $\varphi_f - \sigma \leq b$ —while choice among offers stays quality-only, exactly as fees themselves never sway choice in this model; merit aid is thus a targeted price cut to the bright, and its work is done at the gate. In the scheme there is no price to net against, so the stipend is a side payment entering preferences: a qualifying student ranks the paying college σ utility points higher (one point per thousand), cash being cash even where fees are not. The institutional reading is the best line the instrument owns: when the regulator confiscates the price, price competition does not die—it reappears as side payments, as it does in English bursaries under the fee cap, German Deutschlandstipendien, and the stipend wars of nominally free doctoral programmes. Means-blind admission is not money-blind students.

The post (second edition). Beyond the shocks, the player—and only the player—receives dilemmas: roughly a dozen distinct cards over the library, offered with probability 0.3 a year from year two, each at most once per game. A card names a situation (a rival’s professor is approachable; a donor offers money with conditions attached; an insurer offers a rider against the storm season), prints its price and its odds, and demands a decision before the year proceeds. Cards are drawn on the same separate random stream as the shocks, with all draws—including the success roll—consumed unconditionally, so no decision the player takes can perturb the applicant stream or the rivals’ luck. An unanswered card resolves as a decline, which for the insurance-shaped cards means the gamble is taken by default: an unattended college buys no cover. Card prices are set against the investment technology $\gamma\sqrt{I}$, so that most cards are close to fair at a heavy investor’s margins, poaching carries a zero-sum premium, and one card is priced deliberately below the technology. The reader who checks the arithmetic keeps their money.

The Founders’ Reckoning (second edition). The game’s final table converts remaining endowments into league points: each college is credited $\min(\max(0, E)/15, 8)$ points, announced in the rules from year one and identical in both worlds. The design intent and the honest economics are worth separating. The intent, proposed by the playtest, was a spend-now-versus-save choice. The economics is that for the player the credit is dominated at any realistic scale: a final-round pound invested buys more quality than a pound reckoned until investment approaches five hundred per field, which no solvent college reaches. The Reckoning’s bite lies elsewhere. The rival incumbents end typical games with hoards north of a thousand—capped, those hoards are now worth up to eight points against a challenger, in both worlds. The Reckoning is a difficulty device wearing a savings product’s clothes, and these notes say so plainly.

Events. Each year each solvent college suffers an event with probability $\frac{1}{3}$, drawn from a weighted table of thirty-one shocks: money shocks between -26 and $+32$, quality shocks between -3 and $+4$. Events are drawn from a random-number stream separate from the one generating applicants, so the two sources of luck never contaminate one another; money losses are truncated so that no event pushes an endowment below 5. Events fire before the year’s league table is computed. The morning’s news is news.

Endowments. Starting endowments are world-specific, and deliberately so; they are also private, so the public starting position—the quality table—is identical across worlds. The market’s bottom college holds a war chest (Table 4): the board recapitalises a turnaround, and the chest finances patient learning of the demand curve against the rent. Under the scheme the same chest would be poison to the calibration rather than to the college: with demand guaranteed by the clearing house, early over-spending becomes a winning strategy for the diligent and the asleep alike. The value of a war chest depends on the institution it is spent in.

3 The first institution: the market

In the market world a university chooses, for each field, a fee $\varphi_f \geq 0$ and an admission threshold τ_f . Clearing is simultaneous and mechanical. Every applicant applies to every university whose fee for her field she can afford ($\varphi_f \leq b$). Every university makes an offer to every applicant to a field whose score meets that field’s bar ($s \geq \tau_f$). Every applicant holding at least one offer enrolls at the offer maximising $T_f + \theta R_f$, ties broken at random;

applicants without offers exit without consequence. Offers are unconditional: a university takes every student who accepts. Each enrollee beyond the eighth in a department costs $c = 24$ —roughly three times a typical fee—so oversubscription is not a windfall but an accident, and occasionally an obituary. The fee cap sits at 25, above c , so overflow-for-profit is available in principle; a probe strategy built to try it (price just above c , bar on the floor, never ration) went bankrupt in over ninety per cent of seeds at every cost configuration tested and harvested essentially no overflow. To profit from a crowd one must first be crowded, and by the time a college is attractive enough to overflow at will, maintaining the quality that made it attractive has eaten the margin. The temptation exists; the customers do not.

Two features make this a game rather than an exercise. First, the demand curve is hidden: the player sees the cohort’s score statistics but never its budgets, and must learn by experiment what the market will bear. The bottom college’s central discovery is that it is a local monopolist over a captive segment—the students every rival rejects or prices out—and that the revenue-maximising fee for captives is interior: pricing at the floor buys almost no additional demand. Second, the accepted-offer rule creates a tail risk with a memorable signature. An offer book calibrated to a thirty per cent yield converts at ninety when a quality crossing redirects student choice, and the overage bill from one good year can end the college. The scripted sharp player forecasts its yield from the public league table and rations admission the moment it becomes attractive; human players learn the same lesson, usually the expensive way.

The stipend instrument of Section 2 has a market-specific corollary worth recording: since it works only through affordability, it does nothing for a college whose fee is already within everyone’s means. Merit aid in this market is an instrument of the expensive—a high list price for the captive segment, discounted at the gate for the bright—which is to say, second-degree price discrimination, and which is also to say, the American model. The bottom college, pricing low from necessity, has no use for it until it has a list price worth discounting.

One remark for the theorist. The market’s clearing is itself a single round of student-proposing acceptance—students take their best offer—without quotas and without the right to say no. Seen this way, the two worlds differ in exactly two rights: the right to price, and the right to refuse.

4 The second institution: the scheme

In the scheme world the fee is fixed at $\bar{\varphi} = 5.5$ for every field and every college, admission is means-blind, and budgets are irrelevant. A university chooses, for each field, a quota $q_f \in \{0, 1, \dots, 8\}$ and a threshold τ_f . The seat costs of Section 2 apply unchanged—the rent falls on all eight seats regardless—so the quota is a pure report to the clearing house: it costs nothing to state and binds only the match.

Matriculation is decided by student-proposing deferred acceptance (Gale and Shapley, 1962), run independently for each field. Students rank the four departments of their field strictly by $T_f + \theta R_f$, with vanishing random perturbations guaranteeing strictness. Departments hold responsive preferences: they rank students strictly by score, treat students below τ_f as unacceptable, and hold at most q_f . The algorithm is the standard one. Unmatched students propose down their lists; a department holds the best acceptable proposals up to its quota and rejects the rest, displacing previously held students where necessary; nothing is final until no rejected student has anywhere left to propose. The outcome is the student-optimal stable matching, unique under strict preferences and independent of the or-

der in which proposals are processed (McVitie and Wilson, 1971). The implementation was audited mechanically: across three thousand randomly generated instances the produced matchings contain no blocking pair of either kind—no student preferring a department with a free acceptable seat, and no justified envy—and across twelve hundred simulated game rounds no quota is ever exceeded.

Two theoretical properties of this design do the game’s work. The mechanism is strategy-proof for students (Roth, 1982)—fitting, since the students are honest automata—but no stable mechanism is strategy-proof for the colleges (Roth, 1985), and this one is manipulable in a specific, profitable direction: by capacities (Sönmez, 1997). A department that reports fewer seats than its demand holds only its best proposers; its intake calibre rises, the κ term feeds its teaching, and better teaching attracts better proposers. Since the rent is owed on the seats either way, the manipulation here is undiluted by economy: shrinking the report buys calibre at the price of forgone fees, exactly as the theory has it. In the first edition the scripted sharp player’s mid-game was precisely this manipulation. In the second edition it no longer is, and the reason is instructive: against rivals who spend their guaranteed income rationally (Section 5), the fees a small quota forgoes are worth more than the calibre it buys, and the calibrated best play is volume-first—every seat declared, the threshold doing the selecting. The manipulation remains available, remains undiluted, and remains, at the margin, a mistake. Whether a manipulation pays is a statement about one’s rivals as much as about the mechanism; a player who discovers both halves has discovered a theorem and its comparative statics.

The scheme also removes the market’s tail risk. Deferred acceptance never over-fills a quota, so there is no overage and no yield lottery; what remains is the rent, which both worlds now share, and which the scheme’s guaranteed demand makes easier to pay. Ruin under the scheme is reserved for the college that cannot fill what it keeps—and, as Section 6 shows, the same guarantee that spares the diligent also spares the asleep.

5 The playtest, and the second edition

The reviewer’s strategy deserves to be stated in full, because its full statement is an indictment: declare every seat, keep a workmanlike threshold, spend the entire budget every year, split it equally across the four qualities. Scripted and run over the five hundred calibration seeds of the first edition, it topped the scheme in 43.2 per cent of them—nine points more often than the sharp player—while in the market the same strategy went bankrupt in 57 per cent of seeds. The market was policing full-throttle spending; the scheme was rewarding it.

The diagnosis has three parts, in ascending order of embarrassment. First, there is no downside in the scheme: deferred acceptance never over-fills, so the only ruin available is self-inflicted. Second, the quality technology is symmetric and concave, so splitting equally is close to optimal allocation and no skill is exercised by it. Third—the part that matters—the rival colleges were carrying precautionary cash floats and conservative spending ceilings calibrated against market risk, in a world whose clearing house guarantees their revenue. Precautionary saving without uncertainty is not caution but confusion, and the reviewer’s strategy was simply the only agent in the room spending like a rational actor. The exploit was not a hole in the game; it was a correct answer the designers had failed to anticipate.

The repair is a package of four levers, each calibrated one at a time against the harness. The first is the rationality correction: scheme-world rivals now hold only a thin float, and their plan ceilings loosen by the factor 1.35—a correction, not a handicap, for the reason just given. The second is the Founders’ Reckoning of Section 2, which hands the incumbents’ hoards a capped terminal value. The third is the scholarship instrument of Section 2, in its

world	strategy	mean score	finishes 1st (%)	improves (%)	stagnates or declines (%)	bankrupt (%)
market	naive	0.24	0.4	23.6	76.4	99.6
	sensible	0.37	2.0	30.0	70.0	10.0
	sharp	1.48	32.0	74.0	26.0	9.4
	rustam	0.81	12.4	50.6	49.4	61.0
scheme	naive	0.43	2.0	33.8	66.2	65.6
	sensible	0.46	0.0	43.4	56.6	8.6
	sharp	2.17	33.4	91.8	8.2	1.2
	rustam	1.09	9.8	52.0	48.0	10.6

Table 1: Calibration over 500 seeded histories per cell, player at the worst start (rank 4). Score = initial rank – final rank; “improves” is score ≥ 1 . Second-edition engine throughout; the first edition’s numbers appear in Section 5 where they differ instructively.

per-head form, whose probe evidence is worth one sentence each way: a probe promising a stipend of 10 at a bar of 50 on a sensible budget—nearly every matriculant qualifies—goes bankrupt in effectively every seed in both worlds, so the instrument cannot be spammed, and the new route to ruin it opens is real; while used as designed, at a high bar in a summit fight, its effect is what merit aid’s should be—decisive only when the fighting colleges sit within a stipend’s width of one another in the students’ eyes. The fourth is the card deck, whose accept-everything probe does no better than sensible play once its extra bankruptcies are counted.

The levers interact with best play, and the sharp player had to be re-derived; Section 4 records the headline, the volume-first turn. What survived calibration is this: the sharp player tops roughly a third of seeds in both worlds, as the original brief demanded; the reviewer’s strategy tops the scheme in 9.8 per cent of seeds and reaches the summit by year five and holds it in none of five hundred; and the harness now carries that strategy permanently, as the row *rustam* in every future report. A game owes its best players its vigilance, and the cheapest way to honour a critic is to name the regression test after him.

6 Scripted play and calibration

Each world is calibrated with four scripted players run from the worst starting position over seeds 1–500. The first three tiers are constructed from the same recipes in both worlds. *Naive* makes static mid-range choices and spreads investment evenly. *Sensible* tracks realised demand with its instruments, maintains its qualities against decay, and keeps a reserve. *Sharp* exploits the institution: in the market, revenue hill-climbing on the captive segment, threshold-farming of the intake effect, yield forecasting and top-end rationing; in the scheme, volume-first admissions with the threshold as the selectivity instrument and high-bar stipends in the summit fight; in both, card purchases whenever liquidity and the printed odds allow—every card but the consultancy’s. The fourth row, *rustam*, is the playtest strategy of Section 5, unchanged and permanent. Table 1 reports outcomes; Table 2 reports derived difficulty metrics on the same seeds.

A caveat belongs up front, stated once. Comparing institutions through algorithm performance confounds how hard the world is with how good the hand-crafted algorithm happens to be in it. The naive and sensible rows are comparable by construction—the same abstract recipe mapped onto each world’s instruments. The sharp rows are world-specific by design; they measure each institution’s exploitability ceiling as probed with comparable

world	strategy	s.d. of score	median escape year	never escapes (%)
market	naive	0.46	2	54
	sensible	0.65	6	64
	sharp	1.19	3	9
	rustam	1.00	2	42
scheme	naive	0.68	10	66
	sensible	0.54	11	56
	sharp	0.80	3	7
	rustam	1.11	8	46

Table 2: Difficulty metrics, same 500 seeds. “Escape” is the first year the player leaves rank 4; medians are over escaping seeds.

effort, and nothing stronger is claimed. A cleaner instrument—one identical meta-optimiser given equal compute in each world—is straightforward within the harness and is left for a later vintage of these notes.

With that caveat, four regularities survive. First, the rent bills the sleeping in both worlds, but only the market executes them: static play goes bankrupt in effectively every market seed (99.6 per cent), while under the scheme a third of static colleges survive the early grind. What has changed since the first edition is the fate of the survivors: they used to top the table in 18 per cent of seeds, and now do so in 2.0—rationally-spending rivals no longer leave the summit to the idle. The institutional lesson stands where it belongs: where there is no price, there is no selection, and the clearing house keeps the asleep alive; it just no longer crowns them. Second, mastery pays everywhere and calm is scheme-specific: sharp play reaches the top in a third of seeds in both worlds (32.0 and 33.4 per cent), but with much less noise under the scheme (s.d. 0.80 against 1.19, bankruptcy 1.2 against 9.4)—deferred acceptance abolishes the market’s yield lottery, and what remains of sharp risk in the market is the accepted-offer rule meeting a quality crossing. Third, the rustam rows measure what full-throttle spending is worth under each institution: in the market it is a coin toss against ruin (bankrupt in 61 per cent of seeds), under the scheme it is safe, mediocre, and no longer sovereign (tops 9.8 per cent, reaches the summit by year five and holds it in none). Fourth, prudence buys survival, not distinction, and more than ever: sensible improves in 30 and 43 per cent of seeds but tops in 2.0 and 0.0—the Reckoning hands the hoarding incumbents up to eight free points, and the reserve that keeps a sensible college solvent is precisely what it declines to spend on catching them.

7 Discussion

The calibration is a designer’s equilibrium, not nature’s. The opponents are heuristics, tuned until the game’s difficulty targets held; the numbers in Tables 1 and 2 characterise THE LEAGUE, not British higher education. The reader should know exactly what is, and is not, being claimed: the game does not adjudicate between the institutions, and its parameters were chosen for play, not estimated from anything.

What the comparison does teach is structural, and it survives the caveat. Remove the price instrument, and competition does not disappear—it migrates entirely into quality, where incumbents with hoarded endowments hold the high ground; the resulting reaction functions deter the half-hearted challenges that the market, indifferently, lets through, yet they cannot tell a compounding gambler from a compounding talent. Guarantee demand through a clearing house, and the fixed cost of seats stops selecting: the market forecloses

on the passive, the scheme merely invoices them—and whether the invoiced ever reach the top turns out to depend not on the mechanism but on the conduct of their rivals, which is the second edition’s whole story in one clause. And the endowment experiment deserves its own line: the same war chest that finances patient learning under price competition finances indiscriminate expansion under guaranteed demand—which is why the two worlds start with different money and identical published tables. What capital is worth depends on the institution it is spent in.

The playtest adds a lesson about design itself. The first edition’s rivals were prudent everywhere because prudence had been calibrated where it was needed—the market—and exported, unexamined, to a world that punishes it. The exploit that resulted was not a bug in any line of code; every component behaved as specified. It was an equilibrium the designers had not solved for, found within days by a player who had. The repair, accordingly, was not a patch but a rationality correction, and the test that guards it is the exploiting strategy itself, named for its discoverer and run on every build. Designed economies fail the way real ones do: through agents behaving reasonably in ways the regulator did not model.

For the classroom the game has one further use. Institutions are usually taught as rules and compared as theorems; here they are compared as lived experience, twenty minutes a world. A student who has died of overage understands the accepted-offer rule differently thereafter; a student who has paid three years’ rent on empty seats holds views on the cost of ambition; a student who has tried the Sönmez manipulation against rational rivals, and lost money doing it, has learned that a theorem about mechanisms is not yet a theorem about opponents; and a student who declines the consultancy’s card after checking its odds against $\gamma\sqrt{T}$ has performed a cost–benefit analysis under a deadline, which is more than most syllabi manage. The theorems, of course, were always there. The game merely arranges for them to be discovered in anger.

A Parameters and starting states

B The opponents

The three rivals run fixed personalities under exactly the player’s rules and information. *Harkness*, the balanced incumbent, steers each quality toward a fixed target slightly above its starting level and otherwise hoards; in the scheme it defends the summit with a reaction function in quality: when the published gap to the college below narrows, its target tracks the challenger’s total plus a margin—but the plan, defensive or not, is capped at 46 a year, so a challenger who can outspend the cap can outrun the defence. *Wexford*, the prestige chaser, prices high (or, in the scheme, reports a small quota at a high bar), overweights research roughly two to one, and rescues its teaching only when collapse is visible; in the scheme it opens its war chest when displaced in the table. *Millbrook*, the cash cow, prices low and fills every seat it can, invests a capped upkeep budget, and banks the rest; in the scheme it counter-invests from the hoard when overtaken or when a challenger closes to within four points. Since the second edition all three, in the scheme only, carry the rationality correction of Section 5: cash floats of 8 replace the market-bred reserves of 15–20, and every spending ceiling above scales by 1.35, so the balanced incumbent’s defensive plan runs to 62 a year. The scheme-only reaction functions exist because price competition does not: without them, any adaptive challenger strolled to the top; with them, and with the caps, the scheme blocks the middle of the skill distribution while remaining beatable from its top. All heuristics condition only on public information and their own books. Choice cards are the player’s alone; the rivals’ luck is confined to the shocks.

symbol	value	role	world
T	20	years per game	both
n	40	applicants per year	both
—	8	department capacity (cheap seats)	both
ρ	0.75	annual rent per capacity seat, filled or not	both
c	24	cost per enrollee beyond capacity	marke
—	25	fee cap (equals the budget truncation point)	marke
$\bar{\varphi}$	5.5	regulated fee	schem
r	0.05	interest on unspent funds	both
δ	0.87	quality retention	both
γ	1.5	investment productivity, $\gamma\sqrt{I}$	both
κ	0.14	intake effect on teaching	both
s	$N(60, 13^2)$ on $[20, 100]$	school score	both
b	$10 + 0.12(s-60) + N(0, 3.8^2)$ on $[3, 25]$	budget	marke
θ	$U[0, 1.2]$	research taste	both
p_t	drift $0.02 \tanh(\cdot/10)$ on $[0.3, 0.7]$	STEM share	both
—	$\frac{1}{3}$; weighted table	event probability; event law	both
—	5	endowment floor under event losses	both
—	20; $[40, 95]$; 1.0	stipend cap; bar range (default 70); scheme pull per £1k	both
—	15; 8	Reckoning; endowment per league point; point cap	both
—	0.3; 2	choice-card offer probability; first year offered	both
—	8; 1.35	scheme AI cash float; scheme AI ceiling multiplier	schem

Table 3: The full parameter set, as shipped in `DEFAULT_PARAMS` inside the game file.

university	E (market)	E (scheme)	R_S	T_S	R_H	T_H	sum
Harkness University	120	120	31	30	28	29	118
Wexford Institute	100	100	36	20	28	17	101
Millbrook Metropolitan	130	95	18	24	16	25	83
Greyfriars College	145	105	13	16	11	15	55

Table 4: Starting states. Endowments are private and world-specific (Section 2); the published table—the qualities—is identical across worlds. The player is intended to take Greyfriars; the scoring quietly punishes a softer choice, since score is capped by starting rank minus one.

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